03 — The Control Problem and the Dynamic Programming Principle

Goal of this note. Set up a stochastic optimal control problem precisely: the controlled SDE, the admissible controls, the reward, and the value function. Then state and prove (with a careful sketch) the **dynamic programming principle (DPP)**. It is the controlled version of note 02 eq. (3.1):

Note 02 (no control)	This note (control)
$u(t,x) = \mathbb{E}_{t,x}\big[u(t+h, X_{t+h})\big]$ — eq. (3.1)	$v(t,x) = \sup_\alpha \mathbb{E}_{t,x}\Big[\int_t^{t+h} f\,ds + v(t+h, X^\alpha_{t+h})\Big]$ — eq. (2.1)
$u(s, X_s)$ is a martingale	$\int_t^s f\,dr + v(s, X^\alpha_s)$ is a supermartingale for every α, a martingale for the optimal α

In note 04 the DPP plus Itô's lemma gives the HJB equation, exactly as note 02 §3 got the linear PDE from (3.1).

Prerequisites. Note 01: Itô isometry (3.2), SDEs (Def. 4.2), Itô's lemma (5.1)–(5.2), jumps (6.2)–(6.4). Note 02: Feynman–Kac (1.3), the uncontrolled DPP (3.1).

Notation

Symbol	Meaning	Type
T	fixed horizon	> 0
$(\Omega, \mathcal{F}, \{\mathcal{F}_s\}, \mathbb{P})$	filtered probability space carrying W	—
W_s	m-dimensional Brownian motion	$\mathbb{R}^m$-valued process
A	control set: the values a control may take	closed subset of $\mathbb{R}^k$
a	a generic point of A (a control *value*)	$\in A$
$\alpha = (\alpha_s)_{s\in[t,T]}$	a control *process*	A-valued process
$\mathcal{A}_t$	admissible controls on $[t,T]$ (Def. 1.1)	set of processes
$X_s = X_s^{t,x,\alpha}$	controlled state, $X_t = x$	$\mathbb{R}^d$-valued process
$b(s,x,a)$, $\sigma(s,x,a)$	controlled drift ($\mathbb{R}^d$) and volatility ($\mathbb{R}^{d\times m}$)	functions
K	Lipschitz / growth constant in (H1)–(H3)	> 0
$f(s,x,a)$	running reward rate	function, real

Symbol	Meaning	Type
$g(x)$	terminal reward	function, real
$J(t,x;\alpha)$	expected reward of control α from (t,x)	real
$v(t,x)$	value function $=\sup_\alpha J(t,x;\alpha)$	function
h	DPP step, $0 \le h \le T-t$	time
θ	intermediate time ($=t+h$)	time
S_s^α	$\int_t^s f\,dr + v(s, X_s^\alpha)$ (Cor. 2.2)	process
$\hat{a}(s,x)$	feedback (Markov) control rule	function $\to A$
c	terminal penalty in the example of §4	>0
$\tau = T-t$	time to maturity	≥ 0
$P(t)$, $K(t)$	coefficients of the value function in §4	functions of t
N, $h=\tau/N$, t_n	number of decision dates, step, dates $t_n = t+nh$ in §4	—

Vectors are column vectors; $|\cdot|$ is the Euclidean norm (Frobenius norm for matrices).

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1. Setup & definitions

1.1 The controlled SDE

Roadmap. A control problem has four ingredients: dynamics that depend on a control, a class of allowed controls, a reward, and the value (best achievable reward). We define them in this order.

Fix a starting time $t \in [0,T]$ and state $x \in \mathbb{R}^d$. Given a control process α, the state solves

$$dX_s = b(s, X_s, \alpha_s)\,ds + \sigma(s, X_s, \alpha_s)\,dW_s, \qquad s \in [t,T], \qquad X_t = x. \tag{1.1}$$

For a fixed path of α, (1.1) is an SDE as in note 01 Def. 4.2, but with *random* coefficients $\tilde{b}(s,x,\omega) = b(s,x,\alpha_s(\omega))$. The control enters only through these coefficients.

Assumptions on the coefficients. There is a constant K such that for all $s \in [0,T]$, $x, y \in \mathbb{R}^d$, $a \in A$:

- **(H1) Lipschitz in x, uniformly in a:** $|b(s,x,a) - b(s,y,a)| + |\sigma(s,x,a) - \sigma(s,y,a)| \le K|x-y|$.
- **(H2) Linear growth:** $|b(s,x,a)| + |\sigma(s,x,a)| \le K(1+|x|+|a|)$; and b, σ are continuous in (s,x,a).

“Uniformly in a” is the point of (H1): the same constant works whatever control is applied. This is what makes the estimates below independent of the control.

1.2 Admissible controls

A control must be chosen using only past and present information, exactly like the integrand of the Itô integral (note 01 Def. 3.1).

Definition 1.1 (Admissible control). $\mathcal{A}_t$ is the set of processes $\alpha : [t,T] \times \Omega \to A$ that are

1. **progressively measurable:** for every s, the map $(r,\omega) \mapsto \alpha_r(\omega)$ on $[t,s] \times \Omega$ is $\mathcal{B}([t,s]) \otimes \mathcal{F}_s$ -measurable. (This is "adapted" plus joint measurability in time, which is what is needed for $\int \alpha_r\, dr$ and $\int \sigma(r, X_r, \alpha_r)\, dW_r$ to be adapted. Every adapted process with left- or right-continuous paths qualifies.)
2. **square integrable:**

$$\mathbb{E} \int_t^T |\alpha_s|^2 \, ds < \infty. \tag{1.2}$$

Under (H1)–(H2) and (1.2), Picard iteration (note 01 Def. 4.2) works verbatim with the random coefficients, so (1.1) has a unique solution $X^{t,x,\alpha}$.

Two important subclasses:

- **Feedback (Markov) control:** $\alpha_s = \hat{a}(s, X_s)$ for a measurable function $\hat{a} : [t,T] \times \mathbb{R}^d \to A$. Then (1.1) becomes the *closed-loop* SDE

$$dX_s = b\big(s, X_s, \hat{a}(s,X_s)\big)ds + \sigma\big(s, X_s, \hat{a}(s,X_s)\big)dW_s. \tag{1.3}$$

  It is admissible if (1.3) has a solution and (1.2) holds. This is **not automatic**: if $\hat{a}$ is discontinuous in x, the coefficients of (1.3) are not Lipschitz.
- **Open-loop deterministic control:** $\alpha_s = a(s)$, a deterministic function of time. It cannot react to the noise.

> **Intuition:** a feedback control is a *rule* ("if inventory is q, quote $\delta(q)$"); an open-loop control is a *schedule* fixed in advance. §4 computes how much the rule beats the schedule in a concrete example.

1.3 Two estimates on the controlled state

Roadmap. Every later argument needs (i) X has finite second moments and (ii) X depends continuously on its starting point, *with constants that do not depend on the control*. Both follow from Itô's isometry, Doob's inequality and Gronwall.

Two tools we use:

- **Doob's L^2 inequality.** For a square-integrable martingale M on $[t,T]$, $\mathbb{E}\big[\sup_{t\le r\le s} |M_r|^2\big] \le 4\,\mathbb{E}|M_s|^2$.
- **Gronwall's lemma.** If $\psi \ge 0$ is bounded and $\psi(s) \le a + b\int_t^s \psi(r)\,dr$ on $[t,T]$, then $\psi(s) \le a\, e^{b(s-t)}$.

Lemma 1.2 (moments and stability). Assume (H1)–(H2) and let $\alpha \in \mathcal{A}_t$.

a. There is C depending only on K, T with

$$\mathbb{E}\Big[\sup_{t\le s\le T}|X_s^{t,x,\alpha}|^2\Big] \le C\Big(1+|x|^2+\mathbb{E}\int_t^T|\alpha_s|^2ds\Big). \tag{1.4}$$

b. For two starting points x, y and the **same** α,

$$\mathbb{E}\Big[\sup_{t\le s\le T}\big|X_s^{t,x,\alpha}-X_s^{t,y,\alpha}\big|^2\Big] \le C_1|x-y|^2, \qquad C_1 = 3\,e^{3K^2(T+4)T}. \tag{1.5}$$

C_1 depends only on K and T, **not on** α.

Proof of (b). Let $\Delta_s = X_s^{t,x,\alpha}-X_s^{t,y,\alpha}$ and $\psi(s)=\mathbb{E}\sup_{t\le r\le s}|\Delta_r|^2$. Subtracting the two integral equations,

$$\Delta_s = (x-y)+\int_t^s\big[b(r,X_r^x,\alpha_r)-b(r,X_r^y,\alpha_r)\big]dr+\int_t^s\big[\sigma(r,X_r^x,\alpha_r)-\sigma(r,X_r^y,\alpha_r)\big]dW_r.$$

Use $|p+q+w|^2\le 3(|p|^2+|q|^2+|w|^2)$ and bound the three terms:

- the first is $|x-y|^2$;
- drift term: by Cauchy–Schwarz, $\big|\int_t^s\cdots dr\big|^2\le(s-t)\int_t^s|\cdots|^2dr$, and by (H1) $|\cdots|\le K|\Delta_r|$. Its $\mathbb{E}\sup$ is at most $TK^2\int_t^s\psi(r)\,dr$;
- noise term: Doob's inequality, then Itô's isometry (note 01 eq. 3.2), then (H1): $\mathbb{E}\sup|\int\cdots dW|^2\le 4\,\mathbb{E}\int_t^s|\cdots|^2dr\le 4K^2\int_t^s\psi(r)\,dr$.

Hence $\psi(s)\le 3|x-y|^2+3K^2(T+4)\int_t^s\psi(r)\,dr$, and Gronwall gives (1.5). The control appeared only inside $b(\cdot,\cdot,\alpha_r)$ and $\sigma(\cdot,\cdot,\alpha_r)$, where (H1) removed it.

Proof of (a) is identical with Δ replaced by X, using (H2) instead of (H1); the $|\alpha|^2$ term from (H2) produces $\mathbb{E}\int|\alpha|^2$. (Strictly, Gronwall needs ψ bounded; one first stops X when $|X|$ exceeds n and lets $n\to\infty$.) □

Recap. Moments are finite for every admissible control, and the flow $x\mapsto X^{t,x,\alpha}$ is Lipschitz in mean square, uniformly over controls.

1.4 Reward and value function

Assumption (H3). $f:[0,T]\times\mathbb{R}^d\times A\to\mathbb{R}$ and $g:\mathbb{R}^d\to\mathbb{R}$ are continuous, Lipschitz in x with constant K uniformly in (s,a), and $|f(s,x,a)|+|g(x)|\le K(1+|x|)$.

Definition 1.3 (Reward and value). For $\alpha\in\mathcal{A}_t$,

$$J(t,x;\alpha)=\mathbb{E}\Big[\int_t^T f\big(s,X_s^{t,x,\alpha},\alpha_s\big)\,ds+g\big(X_T^{t,x,\alpha}\big)\Big], \tag{1.6}$$

$$v(t,x)=\sup_{\alpha\in\mathcal{A}_t}J(t,x;\alpha). \tag{1.7}$$

By (H3) and (1.4), J is finite. We assume v is finite (true e.g. if f,g are bounded above).

Comments on the definition:

- v is a **deterministic function** of the starting point, like u in note 02 (Pitfall 3 there). It is defined by an optimization over *processes*, i.e. over an infinite-dimensional set.

- At $t = T$ there is nothing to choose: $v(T, x) = g(x)$.
- **Minimization** (costs) is the same problem with f, g replaced by $-f, -g$. **Discounting** at rate r is included by replacing f with $e^{-r(s-t)} f$ and g with $e^{-r(T-t)} g$, or by adding the discount factor as an extra state as in note 02 eq. (2.2).
- A control α^* is **optimal** if $J(t, x; \alpha^*) = v(t, x)$. It need not exist; ε-optimal controls ($J \geq v - \varepsilon$) always exist by the definition of sup.

1.5 The freezing lemma

One probabilistic tool is used repeatedly below and in note 04.

Lemma 1.4 (Freezing lemma). Let $\mathcal{G} \subseteq \mathcal{F}$ be a σ-algebra, Y a $\mathcal{G}$-measurable random variable, ξ a random variable independent of $\mathcal{G}$, and Φ measurable with $\mathbb{E}|\Phi(Y, \xi)| < \infty$. Then

$$\mathbb{E}\big[\Phi(Y, \xi) \mid \mathcal{G}\big] = \varphi(Y), \qquad \varphi(y) := \mathbb{E}\big[\Phi(y, \xi)\big]. \tag{1.8}$$

In words: condition on what you know by **freezing** it at its realized value, then average over the independent noise. (Proof: true for $\Phi(y, z) = 1_B(y) 1_D(z)$ by independence, then by a monotone class argument.) Note 02's Markov-property step behind eq. (3.1) is an instance of this lemma.

1.6 Extension: controlled jumps

For market making (notes 12–13), the control changes the **intensity** of Poisson-type fills rather than the drift. The state equation becomes

$$dX_s = b(s, X_s, \alpha_s)\, ds + \sigma(s, X_s, \alpha_s)\, dW_s + \sum_{i=1}^{I} \eta_i(s, X_{s-}, \alpha_s)\, dN_s^i, \tag{1.9}$$

where N^i are counting processes whose **intensity** is $\lambda_i(s, X_{s-}, \alpha_s)$: by definition, $N_s^i - \int_t^s \lambda_i(r, X_{r-}, \alpha_r)\, dr$ is a martingale. This generalizes note 01 §6, where λ was a constant. For this to make sense the control must be **predictable** (roughly: determined by information strictly before s, e.g. left-continuous and adapted), so a quote cannot be changed "at the instant of the fill" to exploit it.

Constructing such N^i (by thinning a Poisson random measure, or by a change of measure) is standard but not done here. With bounded intensities and bounded jump sizes η_i, Lemma 1.2 and §2–§3 go through with the obvious changes (the compensated jump integrals $\int \cdots (dN^i - \lambda_i\, ds)$ are handled like the Itô integral, with Itô's isometry replaced by $\mathbb{E}\big(\int H\,(dN - \lambda\, ds)\big)^2 = \mathbb{E}\int H^2 \lambda\, ds$ for predictable H); note 04 §3.4 writes out the jump HJB.

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2. Statement

Theorem 2.1 (Dynamic programming principle). Assume (H1)–(H3) and

- **(H4)** A is compact.

Then for every $(t,x) \in [0,T] \times \mathbb{R}^d$ and every $h \in [0, T-t]$,

$$\boxed{v(t,x) = \sup_{\alpha \in \mathcal{A}_t} \mathbb{E}\Big[\int_t^{t+h} f\big(s, X_s^{t,x,\alpha}, \alpha_s\big)\, ds + v\big(t+h, X_{t+h}^{t,x,\alpha}\big)\Big].} \tag{2.1}$$

Equivalently, the following two inequalities hold:

$$v(t,x) \le \sup_{\alpha \in \mathcal{A}_t} \mathbb{E}\Big[\int_t^{t+h} f\, ds + v(t+h, X_{t+h}^{\alpha})\Big], \tag{2.2}$$

$$\mathbb{E}\Big[\int_t^{t+h} f\, ds + v(t+h, X_{t+h}^{\alpha})\Big] \le v(t,x) \qquad \text{for every } \alpha \in \mathcal{A}_t. \tag{2.3}$$

How to read it. Optimizing over $[t,T]$ in one go is the same as: optimize over $[t, t+h]$, *assuming you will behave optimally afterwards*, so that the future is worth $v(t+h,\cdot)$. The infinite-dimensional problem is split into a short-horizon problem plus a known "continuation value". This is **Bellman's principle**: the tail of an optimal policy is optimal for the tail problem.

How note 02's DPP gains a sup. If $A = \{a_0\}$ is a single point, then $\mathcal{A}_t$ contains only the constant control, $J = v$ is the Feynman–Kac expectation of note 02 eq. (1.3) (with $r = 0$), the sup in (2.1) is over one element, and (2.1) is note 02 eq. (3.1) (with the running reward f added). With more than one control, (2.1) keeps the same tower-property skeleton, and the sup appears because the controller re-optimizes at time $t+h$ knowing X_{t+h}.

Corollary 2.2 (martingale optimality principle). For $\alpha \in \mathcal{A}_t$, define

$$S_s^{\alpha} = \int_t^s f\big(r, X_r^{\alpha}, \alpha_r\big)\, dr + v\big(s, X_s^{\alpha}\big), \qquad s \in [t,T]. \tag{2.4}$$

Then (i) S^{α} is a **supermartingale** for every α; (ii) α is optimal **iff** S^{α} is a **martingale**.

Proof. (i) The conditional version of (2.3), started at $\big(s, X_s^{\alpha}\big)$ instead of (t,x) (same proof as §3.4, conditioned on $\mathcal{F}_s$), gives $\mathbb{E}[S_{s'}^{\alpha} \mid \mathcal{F}_s] \le S_s^{\alpha}$ for $s \le s'$.

ii. Note $S_t^{\alpha} = v(t,x)$ and, since $v(T,\cdot) = g$, $\mathbb{E}[S_T^{\alpha}] = J(t,x;\alpha)$. If S^{α} is a martingale, $J(t,x;\alpha) = \mathbb{E} S_T^{\alpha} = S_t^{\alpha} = v(t,x)$, so α is optimal. Conversely, if α is optimal then $\mathbb{E} S_T^{\alpha} = S_t^{\alpha}$. Fix s and let $D = S_s^{\alpha} - \mathbb{E}[S_T^{\alpha} \mid \mathcal{F}_s]$. By (i), $D \ge 0$. By (i) again, $\mathbb{E} S_s^{\alpha} \le S_t^{\alpha} = \mathbb{E} S_T^{\alpha}$, so $\mathbb{E} D \le 0$. A nonnegative variable with nonpositive mean is 0 a.s., so $S_s^{\alpha} = \mathbb{E}[S_T^{\alpha} \mid \mathcal{F}_s]$: a martingale. □

> **Intuition:** "value so far + value to go" can only decrease in expectation under a bad decision, and stays flat under optimal decisions. Every unit of expected decrease is value lost to a suboptimal action. Compare note 02: there, with no choices, $D\,u + \int D f$ was always a martingale.

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3. Derivation (proof sketch of Theorem 2.1)

Roadmap. (2.2) is the "easy" half: condition on $\mathcal{F}_{t+h}$ and note that whatever the control does after $t+h$ is at best optimal. (2.3) is the "hard" half: we must *build* a control that follows α until $t+h$ and then behaves almost optimally from wherever it lands. The difficulty is choosing the

continuation measurably in the landing point; we avoid it with a countable partition and the continuity of v (Lemma 3.1). One measure-theoretic fact (Fact 3.2) is quoted, not proved.

Throughout, $\theta = t + h$.

3.1 Continuity of the value function

Lemma 3.1. Under (H1)–(H3), with $C_2 = K(T+1)\sqrt{C_1}$,

$$|J(t,x;\alpha) - J(t,y;\alpha)| \le C_2|x-y| \quad \text{for all } \alpha \in \mathcal{A}_t, \qquad |v(t,x) - v(t,y)| \le C_2|x-y|. \tag{3.1}$$

Proof. With $\Delta = X^{t,x,\alpha} - X^{t,y,\alpha}$ and the Lipschitz bounds of (H3),

$$|J(t,x;\alpha) - J(t,y;\alpha)| \le \mathbb{E}\Big[\int_t^T K|\Delta_s|\,ds + K|\Delta_T|\Big] \le K(T+1)\,\mathbb{E}\Big[\sup_s |\Delta_s|\Big] \le K(T+1)\sqrt{C_1}\,|x-y|,$$

using $\mathbb{E}Z \le (\mathbb{E}Z^2)^{1/2}$ (Cauchy–Schwarz) and (1.5).

For v: for each α, $J(t,x;\alpha) \le J(t,y;\alpha) + C_2|x-y| \le v(t,y) + C_2|x-y|$. Taking $\sup_\alpha$ on the left, $v(t,x) \le v(t,y) + C_2|x-y|$. Swap x and y. $\square$

The key is that C_2 is uniform in α; a supremum of uniformly Lipschitz functions is Lipschitz.

3.2 The conditioning fact

Fact 3.2 (conditioning on $\mathcal{F}_\theta$). Let $\alpha \in \mathcal{A}_t$, $X = X^{t,x,\alpha}$.

a. Almost surely,

$$\mathbb{E}\Big[\int_\theta^T f(s, X_s, \alpha_s)\,ds + g(X_T) \;\Big|\; \mathcal{F}_\theta\Big] \le v(\theta, X_\theta). \tag{3.2}$$

b. The value $v(\theta, y)$ is unchanged if, in (1.7) at time θ, we restrict to controls that are functionals of the **future increments** $(W_s - W_\theta)_{s \ge \theta}$ only (hence independent of $\mathcal{F}_\theta$).

Why it is true. After θ, X solves (1.1) started from X_θ, driven by the increments $W_s - W_\theta$, which are independent of $\mathcal{F}_\theta$ (note 01 Def. 1.4(2)). The control after θ may depend on the past, but conditionally on $\mathcal{F}_\theta$ the past is **frozen** (Lemma 1.4): for a fixed past ω, $s \mapsto \alpha_s$ is a control driven by the future increments, started at the frozen state $X_\theta(\omega)$. Its conditional reward is therefore some $J(\theta, X_\theta(\omega); \alpha^\omega) \le v(\theta, X_\theta(\omega))$. Making "for a fixed past" rigorous needs regular conditional probabilities on the canonical Wiener space; see Pham, Lemma 3.3.1 and Thm 3.3.1, or Fleming–Soner Ch. IV. **This step is quoted, not proved here.**

3.3 Proof of (2.2): "$v \le$ sup"

Fix $\alpha \in \mathcal{A}_t$. Split the reward at θ and apply the tower property:

$$J(t,x;\alpha) = \mathbb{E}\Big[\int_t^\theta f\,ds + \mathbb{E}\Big[\int_\theta^T f\,ds + g(X_T) \;\Big|\; \mathcal{F}_\theta\Big]\Big] \le \mathbb{E}\Big[\int_t^\theta f\,ds + v(\theta, X_\theta)\Big],$$

where the inequality is (3.2). The right side is at most the sup over α in (2.2). Taking $\sup_\alpha$ on the left gives (2.2). $\square$

3.4 Proof of (2.3): "$v \geq$ every candidate"

Roadmap. Fix any $\alpha \in \mathcal{A}_t$ and $\varepsilon, \delta > 0$. We paste: "use α until θ, then switch to an ε-optimal control for the cell of a fine partition in which X_θ landed." This pasted control is admissible, so its reward is at most $v(t,x)$; and its reward is within $\varepsilon + 2C_2\delta$ of the left side of (2.3).

Step 1: discretize the landing point. Partition $\mathbb{R}^d$ into countably many disjoint Borel sets $B_1, B_2, \ldots$ of diameter $< \delta$ (e.g. half-open cubes), and pick $x_i \in B_i$.

Step 2: choose ε-optimal continuations. For each i, by Fact 3.2(b) choose $\alpha^i \in \mathcal{A}_\theta$, a functional of the increments $(W_s - W_\theta)_{s \geq \theta}$, with

$$J(\theta, x_i; \alpha^i) \geq v(\theta, x_i) - \varepsilon.$$

This is a *countable* family of choices, so no measurable-selection theorem is needed.

Step 3: paste. Define

$$\hat{\alpha}_s = \begin{cases} \alpha_s, & t \leq s < \theta, \\ \sum_i \mathbf{1}\{X_\theta^{t,x,\alpha} \in B_i\}\, \alpha^i_s, & \theta \leq s \leq T. \end{cases} \tag{3.3}$$

Admissibility. The event $\{X_\theta \in B_i\}$ is in $\mathcal{F}_\theta$, and each α^i is progressively measurable on $[\theta, T]$, so $\hat{\alpha}$ is progressively measurable. It takes values in A, which is compact by (H4), so (1.2) holds. Hence $\hat{\alpha} \in \mathcal{A}_t$. (Without (H4) one needs an extra truncation; this is where compactness is used.)

Step 4: compute the conditional reward of $\hat{\alpha}$. On $[t, \theta]$, $\hat{\alpha} = \alpha$, so $X^{\hat{\alpha}} = X^\alpha$ there. On the event $\{X_\theta \in B_i\}$, after θ the state solves (1.1) from X_θ with control α^i. Here X_θ is $\mathcal{F}_\theta$-measurable and α^i together with the driving increments is independent of $\mathcal{F}_\theta$. By the freezing lemma (1.8) and pathwise uniqueness for (1.1),

$$\mathbb{E}\Big[\int_\theta^T f\, ds + g(X_T^{\hat{\alpha}}) \,\Big|\, \mathcal{F}_\theta \Big] = J(\theta, X_\theta; \alpha^i) \qquad \text{on } \{X_\theta \in B_i\}.$$

Step 5: compare with v using continuity. On $\{X_\theta \in B_i\}$ we have $|X_\theta - x_i| < \delta$, so by (3.1) twice,

$$J(\theta, X_\theta; \alpha^i) \geq J(\theta, x_i; \alpha^i) - C_2\delta \geq v(\theta, x_i) - \varepsilon - C_2\delta \geq v(\theta, X_\theta) - \varepsilon - 2C_2\delta.$$

This holds on every cell, hence a.s.

Step 6: conclude. Write $i(y)$ for the index of the cell containing y. Since $\hat{\alpha}$ is admissible, by the tower property, Step 4 and Step 5,

$$v(t,x) \geq J(t,x;\hat{\alpha}) = \mathbb{E}\Big[\int_t^\theta f(s, X_s^\alpha, \alpha_s)\, ds + J\big(\theta, X_\theta^\alpha; \alpha^{i(X_\theta^\alpha)}\big) \Big] \geq \mathbb{E}\Big[\int_t^\theta f\, ds + v(\theta, X_\theta^\alpha) \Big] - \varepsilon - 2C_2\delta.$$

ε, δ are arbitrary, so (2.3) holds. $\square$

Recap. The "$\leq$" half is just conditioning. The "$\geq$" half needs (i) the ability to paste controls at θ (admissibility is stable under pasting), and (ii) continuity of v in x (Lemma 3.1), which lets a countable family of near-optimal controls cover all landing points. Without continuity, the DPP still holds in a weaker form (Bouchard–Touzi 2011), but that refinement is not needed here.

> **Remark (stopping times).** The same proof works with θ replaced by any stopping time with values in $[t, T]$. We only use deterministic $\theta = t + h$ in note 04.

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4. Worked example: steering a noisy state to zero

Roadmap. A 1-d problem where the controller pays for effort and is penalized for distance from 0 at T. We compute (i) the best **open-loop** value by elementary calculus, (ii) the best value over controls that decide at N dates, by running the DPP backward in discrete time — this gives an exact formula for every N, and (iii) the limit $N \to \infty$. The gap between (i) and (iii) is the value of using feedback.

This problem violates (H3)–(H4) (quadratic rewards, $A = \mathbb{R}$). Everything in this section is proved directly by elementary arguments, so Theorem 2.1 is not used.

4.1 The problem

State and reward ($d = m = k = 1$, $A = \mathbb{R}$, $c > 0$, $\sigma > 0$):

$$dX_s = \alpha_s\, ds + \sigma\, dW_s, \qquad X_t = x, \tag{4.1}$$

$$J(t, x; \alpha) = \mathbb{E}\Big[- \int_t^T \tfrac{1}{2}\alpha_s^2\, ds - \tfrac{1}{2} c\, X_T^2 \Big]. \tag{4.2}$$

Reading: X is, e.g., an inventory you must bring to 0 by T; α is the trading rate with a quadratic cost; $\sigma\, dW$ is unhedgeable order flow. (Almgren–Chriss, note 11, is a relative.)

4.2 Best open-loop control

For deterministic $\alpha = a(\cdot)$, let $I = \int_t^T a(s)\, ds$. Then from (4.1)

$$X_T = x + I + \sigma(W_T - W_t) \sim \mathcal{N}\big(x + I,\ \sigma^2 \tau\big), \tag{4.3}$$

so $\mathbb{E} X_T^2 = (x + I)^2 + \sigma^2 \tau$ and $J = -\tfrac{1}{2}\int a^2 - \tfrac{1}{2} c\big[(x + I)^2 + \sigma^2 \tau\big]$.

Step 1: fix I, minimize effort. By Cauchy–Schwarz, $I^2 = \big(\int_t^T 1 \cdot a\big)^2 \le \tau \int_t^T a^2$, with equality iff a is constant. So the best schedule with total I is constant, $a \equiv I/\tau$, and $J = -\frac{I^2}{2\tau} - \frac{1}{2} c (x + I)^2 - \frac{1}{2} c \sigma^2 \tau$.

Step 2: optimize I. Setting the derivative to zero, $-I/\tau - c(x + I) = 0$, so $I^* = -\dfrac{c\tau x}{1 + c\tau}$ and $x + I^* = \dfrac{x}{1 + c\tau}$. Then $\dfrac{(I^*)^2}{\tau} + c(x + I^*)^2 = \dfrac{c^2 \tau x^2 + c x^2}{(1 + c\tau)^2} = \dfrac{c x^2}{1 + c\tau}$. Hence

$$v_{\mathrm{OL}}(t, x) = -\frac{c\, x^2}{2(1 + c\tau)} - \frac{1}{2} c \sigma^2 \tau. \tag{4.4}$$

The noise term $-\frac{1}{2} c \sigma^2 \tau$ is the full penalty on the accumulated variance: an open-loop schedule cannot react to it at all.

4.3 Decisions at N dates: the DPP in discrete time

Let $h = \tau/N$, $t_n = t + nh$. Restrict to controls that are constant on each $[t_n, t_{n+1})$, with value a_n chosen using information up to t_n (a_n is $\mathcal{F}_{t_n}$-measurable, $\mathbb{E}a_n^2 < \infty$). These are admissible controls of the continuous problem, so the best reward among them, $v^h(t,x)$, satisfies

$$v^h(t,x) \le v(t,x). \tag{4.5}$$

Writing $X_n = X_{t_n}$ and $\xi_n = (W_{t_{n+1}} - W_{t_n})/\sqrt{h} \sim \mathcal{N}(0,1)$ (independent of $\mathcal{F}_{t_n}$), (4.1) gives exactly $X_{n+1} = X_n + a_n h + \sigma\sqrt{h}\,\xi_n$, and the running reward over a step is exactly $-\tfrac12 a_n^2 h$.

Backward recursion. Define $V_N(y) = -\tfrac12 c y^2$ and, for $n = N-1, \dots, 0$,

$$V_n(y) = \sup_{a \in \mathbb{R}} \Big\{ -\tfrac12 a^2 h + \mathbb{E}\big[V_{n+1}\big(y + ah + \sigma\sqrt{h}\,\xi\big)\big] \Big\}, \qquad \xi \sim \mathcal{N}(0,1). \tag{4.6}$$

This is (2.1) with h one step and the sup over a single number a, because the control is constant over the step.

Solve the recursion. Guess $V_{n+1}(y) = -\tfrac12 P_{n+1} y^2 - K_{n+1}$ (true for $n+1 = N$ with $P_N = c$, $K_N = 0$). Using $\mathbb{E}(m + \sigma\sqrt{h}\xi)^2 = m^2 + \sigma^2 h$,

$$V_n(y) = \sup_a \Big\{ -\tfrac12 a^2 h - \tfrac12 P_{n+1}(y + ah)^2 \Big\} - \tfrac12 \sigma^2 h P_{n+1} - K_{n+1}.$$

The bracket is a concave quadratic in a. Its derivative is $-ah - P_{n+1}h(y + ah)$, which vanishes at

$$a_n^* = -\frac{P_{n+1}}{1 + P_{n+1}h}\, y. \tag{4.7}$$

Then $y + a_n^* h = \dfrac{y}{1 + P_{n+1}h}$ and the bracket equals $-\tfrac12 \dfrac{h P_{n+1}^2 y^2 + P_{n+1} y^2}{(1 + P_{n+1}h)^2} = -\tfrac12 \dfrac{P_{n+1}}{1 + P_{n+1}h} y^2$. So the guess propagates with

$$P_n = \frac{P_{n+1}}{1 + P_{n+1}h}, \qquad K_n = K_{n+1} + \tfrac12 \sigma^2 h\, P_{n+1}. \tag{4.8}$$

Closed form. The first relation is $1/P_n = 1/P_{n+1} + h$, so $1/P_n = 1/c + (N-n)h$:

$$P_n = \frac{c}{1 + c(T - t_n)}, \qquad a_n^* = -P_n\, X_n, \qquad K_0 = \tfrac12 \sigma^2 h \sum_{j=0}^{N-1} \frac{c}{1 + c\, jh}. \tag{4.9}$$

(For a_n^*: $\frac{P_{n+1}}{1+P_{n+1}h} = P_n$ by (4.8). For K_0: $K_0 = \tfrac12\sigma^2 h \sum_{n=1}^{N} P_n$ and $T - t_n = (N-n)h$ runs over jh, $j = 0, \dots, N-1$.)

Why the recursion gives the optimum (discrete verification). Let (a_n) be any admissible discrete control and X_n its state. By the freezing lemma (1.8) with $\mathcal{G} = \mathcal{F}_{t_n}$, $Y = (X_n, a_n)$, and $\xi = \xi_n$:

$$\mathbb{E}\big[-\tfrac12 a_n^2 h + V_{n+1}(X_{n+1}) \mid \mathcal{F}_{t_n} \big] = \Big\{ -\tfrac12 a^2 h + \mathbb{E} V_{n+1}(y + ah + \sigma\sqrt{h}\xi) \Big\}\Big|_{y = X_n,\, a = a_n} \le V_n(X_n),$$

with equality iff $a_n = a_n^*(X_n)$. Take expectations and sum over $n = 0, \dots, N-1$; the V_n telescope:

$$J(t,x;(a_n)) = \mathbb{E}\Big[\sum_{n} -\tfrac{1}{2}a_n^2 h + V_N(X_N)\Big] \le V_0(x),$$

with equality for $a_n = -P_n X_n$. (Integrability: $\mathbb{E}a_n^2 < \infty$ implies $\mathbb{E}X_n^2 < \infty$, so all terms are finite.) Hence $v^h = V_0$:

$$v^h(t,x) = -\tfrac{1}{2}P_0 x^2 - K_0. \tag{4.10}$$

This telescoping argument is the discrete prototype of the verification theorem of note 04, and the inequality with equality iff optimal is Corollary 2.2 in discrete time.

4.4 The continuous-time limit

As $N \to \infty$, $P_0 = c/(1+c\tau)$ does not change, and K_0 is a left Riemann sum of a decreasing function:

$$K_0 \;\longrightarrow\; \tfrac{1}{2}\sigma^2 \int_0^\tau \frac{c}{1+cu}\,du = \tfrac{1}{2}\sigma^2 \ln(1+c\tau),$$

with error $\approx \tfrac{1}{4}\sigma^2 h\left(c - \frac{c}{1+c\tau}\right) = O(h)$ (trapezoid correction). By (4.5), $v \ge \lim_N v^h$, i.e.

$$v(t,x) \;\ge\; -\frac{c\,x^2}{2(1+c\tau)} - \frac{1}{2}\sigma^2 \ln(1+c\tau). \tag{4.11}$$

In note 04 (Example 4.1) the HJB equation plus the verification theorem shows that (4.11) holds with **equality**, with optimal feedback $\alpha_s^* = -P(s)X_s$, $P(s) = c/(1+c(T-s))$. Here we have only the lower bound, because a continuous-time control could in principle do better than every piecewise-constant one.

Value of feedback. Comparing (4.4) and (4.11), the x-dependent parts coincide; the difference is in the noise term:

$$v - v_{\mathrm{OL}} \;\ge\; \tfrac{1}{2}\sigma^2\big[c\tau - \ln(1+c\tau)\big] > 0 \quad (c\tau > 0). \tag{4.12}$$

(The inequality $\ln(1+z) < z$ for $z > 0$.) For $\sigma = 0$ the gap vanishes: without noise there is nothing to react to.

4.5 Numbers

$c = 4$, $\sigma = 1$, $T = 1$, $t = 0$, $x = 1$ (so $\tau = 1$, $P_0 = 0.8$):

Quantity	Formula	Value
open-loop value	(4.4): $-0.4 - 2$	-2.4000
$N = 10$ dates	(4.10)	-1.29102
$N = 100$ dates	(4.10)	-1.21278
$N = 1000$ dates	(4.10)	-1.20552
limit	(4.11): $-0.4 - \tfrac{1}{2}\ln 5$	-1.20472
first action	$a_0^* = -P_0 x$	-0.8

The errors $0.0863,\ 0.0081,\ 0.0008$ decrease like h, and the predicted leading error $\frac{1}{4}h(4-0.8) = 0.8h$ matches. Feedback cuts the expected cost roughly in half ($2.40 \to 1.20$): the open-loop controller pays $\frac{1}{2}c\sigma^2\tau = 2$ for terminal variance, the feedback controller only $\frac{1}{2}\sigma^2 \ln 5 = 0.80$, because it keeps correcting the noise as it arrives.

DPP sanity check (Corollary 2.2). Take $h = 0.25$ and use the limit formula (4.11) as v (justified in note 04). Then $P(0.25) = 4/(1+3) = 1$. - Under $\alpha \equiv 0$ on $[0, 0.25]$: $X_{0.25} = 1 + W_{0.25}$, no running cost, and $\mathbb{E}\, v(0.25, X_{0.25}) = -\frac{1}{2}\cdot 1 \cdot (1 + 0.25) - \frac{1}{2}\ln 4 = -1.31815 < -1.20472 = v(0,1)$: a strict loss of 0.113, as (2.3) requires. - Under the optimal feedback on $[0, 0.25]$, the same quantity equals $v(0,1)$ (martingale). The Monte Carlo test checks both.

> **Intuition:** the gain $P(t) = c/(1 + c(T-t))$ increases toward c as $t \to T$: early on, a deviation can be corrected gradually and cheaply (and more noise is still to come, so effort now is partly wasted); near T the controller pushes hard. The same urgency appears in Almgren–Chriss and in inventory skew near the close.

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5. Numerical check

Script: `code/control_dpp_checks.py`; tests: `code/tests/test_03_control_dpp.py` (run `python -m pytest code/tests -q`).

Check	Methods compared	Expected	Tolerance
Bellman recursion	iterate (4.8) vs closed form (4.9)–(4.10)	$P_0 = 0.8$, v^h for $N = 10, 100, 1000$	10^{-12}
$O(h)$ convergence	$N\,(v^h - v)$ at $N = 1000$ vs predicted leading term	-0.8	2%
Generic DPP on a grid	brute-force $\max_a$ + Gauss–Hermite + spline, **no quadratic ansatz**, vs (4.10), $N = 10$	-1.29102	$5 \cdot 10^{-4}$
Open-loop	BFGS over 20 deterministic control values vs (4.4); optimizer is constant	-2.4	10^{-6}
Feedback value	Monte Carlo of $\alpha = -P(s)X$ vs (4.11)	-1.20472	4 s.e. + Euler bias
Open-loop MC	Monte Carlo of the constant schedule vs (4.4)	-2.4	4 s.e.
DPP / Cor. 2.2	MC of both sides of (2.1) at $h = 0.25$	-1.20472 (optimal), -1.31815 ($\alpha = 0$)	4 s.e. + bias

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6. Pitfalls / intuition

1. v **is defined by an optimization over processes**, not numbers. The DPP is what reduces it to a sequence of finite-dimensional choices; in note 04 it reduces to a pointwise $\sup_a$ in the Hamiltonian.
2. **The "$\geq$" half of the DPP is the hard one.** It needs pasting of controls and continuity (or measurable selection). Proofs that "just condition on $\mathcal{F}_{t+h}$" prove only (2.2).
3. **Admissibility is part of the problem.** Changing $\mathcal{A}_t$ (e.g. allowing unbounded controls without (1.2)) can change v. Feedback rules must produce a well-posed closed-loop SDE (1.3).
4. **Open-loop ≠ feedback under noise.** The best deterministic schedule is generally strictly worse (4.12). Without noise ($\sigma = 0$) they coincide.
5. **Discrete decision dates give lower bounds.** $v^h \leq v$ always (4.5); in this example the gap is $O(h)$. This is the practical meaning of "re-quote more often".
6. **Heuristic vs proved.** Fact 3.2 (conditioning with a frozen past) is quoted from Pham. The discrete-time results of §4.3 are fully proved; the equality in (4.11) is proved in note 04.

中文小结

- 控制问题四要素：受控 SDE $dX = b(s, X, \alpha)ds + \sigma(s, X, \alpha)dW$；容许控制（循序可测、平方可积，只能用当前及过去信息）；收益 $J = \mathbb{E}[\int f\, ds + g(X_T)]$；值函数 $v = \sup_\alpha J$，是起点 (t, x) 的确定性函数，$v(T, x) = g(x)$。
- 两个估计：在关于 x 一致 Lipschitz 的条件下，X 的二阶矩有限，且对初值的依赖是 Lipschitz 的，常数与控制无关；由此 v 关于 x Lipschitz 连续。
- 动态规划原理：$v(t, x) = \sup_\alpha \mathbb{E}[\int_t^{t+h} f\, ds + v(t+h, X_{t+h})]$。它是第 02 篇 (3.1) 加上一个 sup：只有一个控制时退化为无控制 DPP。"$\leq$"方向只需条件期望；"$\geq$"方向需要在 $t+h$ 处拼接 ε-最优控制，靠可数划分和 v 的连续性避免可测选择问题。条件化时"冻结过去"的严格化引自 Pham。
- 鞅最优性原理：$\int f + v(s, X_s)$ 对任意控制是上鞅，当且仅当控制最优时是鞅。
- 例子（把带噪状态推向 $\mathbf{0}$）：离散时间 DPP 可精确求解，$P_n = c/(1 + c\tau_n)$，最优反馈 $a = -P_n X_n$；连续极限 $v \geq -\frac{cx^2}{2(1+c\tau)} - \frac{1}{2}\sigma^2 \ln(1 + c\tau)$（第 04 篇证明取等号）。开环最优值为 $-\frac{cx^2}{2(1+c\tau)} - \frac{1}{2}c\sigma^2\tau$。$c = 4, \sigma = 1, \tau = 1, x = 1$ 时：反馈 -1.205，开环 -2.4，反馈把成本几乎减半；离散决策误差为 $O(h)$。
- 数值验证：Bellman 递推、无假设网格 DPP、开环数值优化、蒙特卡洛（含 DPP 两侧检验）全部与闭式解一致。